PYTHON INTERVIEW PRACTICE

Blocks 1–6 | Verbal Questions, Coding Exercises & Mock Interview

A structured 4–6 hour practice workbook covering Python fundamentals, collections, functions, idiomatic Python, exceptions and file operations.

Hour	Focus
1	Python fundamentals and verbal questions
2	Lists, tuples and strings
3	Dictionaries and sets
4	Functions, scope, comprehensions and generators
5	Exception handling and file operations
6	Output questions, coding test and mock interview

How to use this workbook

• Answer verbal questions aloud in 30–60 seconds.

• Write code before checking references.

• Explain your approach and edge cases.

• For four hours, complete Hours 1–4; reserve Hours 5–6 for a longer session or another day.

Hour 1 – Python Fundamentals

Common verbal interview questions

What is Python?
A high-level, general-purpose language with readable syntax. It supports procedural, object-oriented and functional programming.

Main features?
Readable syntax, dynamic typing, extensive standard library, cross-platform support, multiple paradigms and automatic memory management.

What is dynamic typing?
A name can refer to objects of different types during execution; you do not declare its type in advance.

Mutable vs immutable?
Mutable objects can change after creation. Lists, dictionaries and sets are mutable; strings, integers, floats and tuples are immutable. A tuple can contain mutable objects.

== vs is?
== compares values; is checks object identity.

break, continue, pass?
break exits a loop; continue skips to the next iteration; pass is a no-operation placeholder.

What is type casting?
Converting a value from one type to another, such as int('100').

List vs tuple?
A list is mutable; a tuple is immutable. Use lists for changing collections and tuples for fixed groupings.

Truthy and falsy values?
Falsy examples include False, None, zero, empty strings and empty containers. Most other values are truthy.

for vs while?
for iterates over an iterable; while repeats as long as a condition is true.

Additional questions to answer aloud

• Interpreted vs compiled language?

• append() vs extend()?

• What are slicing and negative indexing?

• remove() vs pop() vs del?

• sort() vs sorted()?

• None vs False vs 0?

• What do enumerate() and zip() do?

Hour 2 – Lists, Tuples and Strings

Coding practice

Reverse a string
'Python' → 'nohtyP'

Find duplicates
[10,20,30,20,40,10,50] → duplicate values 20 and 10

Remove duplicates preserving order
[10,20,10,30,20,40] → [10,20,30,40]

Second-largest distinct number
[10,50,30,70,70,20] → 50

Palindrome check
'madam' → Palindrome

Count vowels
'Python Developer' → 5

Common elements
[10,20,30,40] and [30,40,50,60] → [30,40]

Missing number
[1,2,3,5,6], range 1..6 → 4

Hour 3 – Dictionaries and Sets

Verbal questions

• What is a dictionary? Explain key-value pairs and unique, hashable keys.

• dict.get(key) vs dict[key]?

• What is a set, and how does it remove duplicates?

• Explain union, intersection, difference and symmetric difference.

• What does hashable mean? Give examples.

• Why can't a list be a dictionary key?

Coding practice

Count element frequency
[10,20,10,30,20,10] → {10:3, 20:2, 30:1}

Most frequent character
'programming' → r:2; report ties if applicable

Merge dictionaries
Combine {'name':'Server-01','status':'Active'} and {'ip':'10.0.0.1','region':'India'}.

Common IP addresses
Find intersection of {'10.0.0.1','10.0.0.2','10.0.0.3'} and {'10.0.0.2','10.0.0.3','10.0.0.4'}.

Group servers by status
Group server dictionaries into Active and Inactive name lists.

Find missing keys
Required: name, ip, status, region. Given name and ip; expected missing: status, region.

Hour 4 – Functions and Idiomatic Python

Verbal questions

What is a function?
A reusable block of code that performs a task.

return vs print()?
return passes a value to the caller; print displays it.

*args vs **kwargs?
*args collects extra positional arguments in a tuple; **kwargs collects keyword arguments in a dictionary.

Explain LEGB.
Local, Enclosing, Global and Built-in are variable lookup scopes.

What is lambda?
A small anonymous function expression.

map() vs filter()?
map transforms each item; filter selects items using a predicate.

What is a generator?
A generator uses yield to produce values lazily.

List comprehension vs generator expression?
A list comprehension builds a list immediately; a generator expression produces values on demand.

any() vs all()?
any is true if at least one item is truthy; all is true if every item is truthy.

Coding practice

• Write a function to check even or odd.

• Calculate factorial(5) → 120.

• Write find_max(*args) and call it with 10, 40, 20, 70, 30.

• Sort alert dictionaries by severity descending using sorted(..., key=...).

• Filter High severity alerts using filter() or a comprehension.

• Write a generator that yields even numbers from 0 through N.

• Validate an API payload; raise ValueError if alert_id, severity or source is missing.

Hour 5 – Exception Handling and File Operations

Verbal questions

What is an exception?
An event during execution that interrupts normal program flow.

Syntax error vs exception?
A syntax error violates Python grammar; an exception is raised during execution.

Explain try/except/else/finally.
try contains risky code; except handles matching errors; else runs if no exception; finally runs when leaving the construct.

raise vs assert?
raise explicitly raises an exception. assert checks an assumption and may be disabled under optimization; don't use it for essential input validation.

Why avoid bare except?
It can hide unexpected errors. Catch specific exception types.

What is a context manager?
A resource-management construct used with with; it performs setup and cleanup, such as closing a file.

read vs readline vs readlines?
read returns remaining content as a string; readline returns one line; readlines returns remaining lines as a list.

w vs a?
w creates or truncates a file; a appends to the end.

How do you handle a missing file?
Catch FileNotFoundError around the file-opening operation.

Coding practice

• Handle invalid numeric input and division by zero separately.

• Validate an alert dictionary and raise ValueError for missing required fields.

• Read a log file line by line and count INFO, WARNING and ERROR entries.

• Print only lines containing ERROR.

• Copy source.txt to destination.txt; handle missing source and permission errors.

• Append a timestamped message to automation.log without overwriting earlier content.

• Process a large file line by line and handle FileNotFoundError.

File modes

Mode	Meaning
r	Read an existing file
w	Write; creates or truncates
a	Append; creates if missing
x	Create exclusively; fails if it exists
rb / wb	Read / write binary data
r+	Read and write

Hour 6 – Mixed Mock Interview

10 output-based questions

Question 1: Assignment shares the same list object; predict the output.

```
a=[1,2,3]; b=a; b.append(4); print(a)
```

Question 2: Predict both outputs.

```
numbers=[10,20,30,40,50]; print(numbers[1:4]); print(numbers[::-1])
```

Question 3: Which characters are printed?

```
for char in 'Python':
    if char == 'h': break
    print(char)
```

Question 4: What default is returned?

```
data={'a':10,'b':20}; print(data.get('c',0))
```

Question 5: Which values are selected?

```
numbers=[1,2,3,4,5]; print([x for x in numbers if x%2==0])
```

Question 6: Predict both results.

```
def add(a,b=10): return a+b; print(add(5)); print(add(5,20))
```

Question 7: Explain the order.

```
def test():
    try: return 'try'
    finally: print('finally')
print(test())
```

Question 8: What list is produced?

```
numbers=[1,2,3]; result=map(lambda x:x*2,numbers); print(list(result))
```

Question 9: What does & mean for sets?

```
a={1,2,3}; b={3,4,5}; print(a & b)
```

Question 10: Predict the output order.

```
try: number=int('100')
except ValueError: print('Error')
else: print('Success')
finally: print('Completed')
```

10 mixed coding challenges

• Find the second-largest distinct number in a list.

• Remove duplicates while preserving order.

• Count word frequency in a sentence.

• Find common values in two lists.

• Find the missing number from 1 to N.

• Group servers by status.

• Sort alerts by severity descending.

• Validate an API payload and raise ValueError when invalid.

• Count ERROR messages in a file.

• Create a generator for even numbers up to N.

Practice input data

```
numbers = [10, 20, 50, 30, 70, 70, 20]
list1 = [1, 2, 3, 4, 5]
list2 = [3, 4, 5, 6, 7]
sentence = "python is easy and python is powerful"
servers = [
    {"name": "S1", "status": "Active"},
    {"name": "S2", "status": "Inactive"},
    {"name": "S3", "status": "Active"},
    {"name": "S4", "status": "Inactive"}
]
alerts = [
    {"id": 1, "severity": 2},
    {"id": 2, "severity": 4},
    {"id": 3, "severity": 1},
    {"id": 4, "severity": 3}
]
```

Real-world scenarios

• API response validation: verify status and required alert_id; return the ID or raise a meaningful exception.

• Critical asset monitoring: compare critical_assets=['SERVER-01','SERVER-02','SERVER-03'] with active_assets=['SERVER-01','SERVER-03']; identify missing assets.

• Security alert processing: count alerts by severity, extract High alerts, total them and sort by severity.

• Large log processing: explain line-by-line reading, counting failed logins and handling missing files.

• Automation reliability: explain exception handling, logging and how to isolate one customer's failure so other customers can continue.

20 verbal mock interview questions

• Why did you choose Python?

• What are Python's built-in data types?

• Mutable vs immutable?

• List vs tuple?

• List vs set?

• List vs dictionary?

• What is a dictionary comprehension?

• What is a function?

• What are *args and **kwargs?

• What is lambda?

• What is a generator?

• map() vs filter()?

• What is exception handling?

• What is finally?

• What is a context manager?

• How do you read a large file?

• Shallow copy vs deep copy?

• Explain LEGB.

• == vs is?

• How do you debug a Python program?

Suggested 4–6 hour study plan

Hour	Tasks
1	Verbal fundamentals; explain answers aloud.
2	Solve list, tuple and string problems.
3	Solve dictionary and set problems.
4	Solve function, comprehension and generator problems.
5	Solve exception and file-handling problems.
6	Complete output questions, coding test and mock interview.

Four-hour option: Complete Hours 1–4, then schedule Hours 5–6 separately.

Self-assessment checklist

• I can explain each concept in 30–60 seconds.

• I can write basic collection and string programs without notes.

• I understand dictionary and set operations.

• I can write and call functions, use *args and **kwargs.

• I understand comprehensions, generators, map and filter.

• I can handle specific exceptions and raise ValueError.

• I can read, write and append files safely with with open().

• I can process large files line by line.

• I can explain my code and discuss edge cases.

• I can solve at least 70–80% of the practice problems independently.

Recommended next step: complete this workbook, then do a timed mock interview before starting Block 7 (Object-Oriented Programming).